23/06/26

CASE Statement

- Allows to create a new column based on logical conditions applied on other columns.

- As case statement allows us to classify our numerical data into categorical groupings

Syntax: SELECT column1,
CASE
WHEN condition1 THEN result1
WHEN condition2 THEN result2
ELSE default_result
END AS column_name
FROM table_name

e.g.

Employees

emp_id	Name	Salary
101	John	3000
102	Peter	7000
103	Samson	12000
104	David	4000
105	Goliath	300

want to classify the salary into categories (low salary, medium or high

Salary below 5000 = Low
Salary between 5000 and 10 000 = medium
Salary above 10 000 = high

} business logic

The case statement is placed within the select statement. case statement creates a new column

```
SELECT name,
       Salary
CASE
    WHEN Salary < 5000 THEN 'Low'
    WHEN Salary BETWEEN 5000 AND 10 000
     THEN 'MED'
WHEN Salary > 10 000 THEN 'High'
END AS Salary_bracket
FROM Employees;
```

name	Salary	Salary-bracket
John	3000	Low
Peter	7000	Med
Samson	12000	High
David	4000	Low
Goliath	300	Low

- use table from exercise

```
SELECT Product name
       Price
CASE   WHEN Price > 1000 THEN 'Expensive'
       WHEN Price BETWEEN 100 and 1000
        THEN 'Mid-range'
 WHEN Price < 100 THEN 'Budget'
END AS Price_category
   FROM Products
```

2. Table : Orders

~~Order_id~~

```
SELECT customer_name,
       amount
CASE
    WHEN amount >1000 THEN 'High value'
    WHEN amount BETWEEN 500-999,99
     THEN 'Med'
    WHEN amount <500 THEN 'Low'
```

cust-name	amount	order-value_category
~~Order_id~~	~~cust_name~~	~~amount~~
Alice	~~Alice~~ 150	Low value
Bob	560	med
Charlie	999,99	med
Diana	45,50	Low
Ethan	1200	High

3. Table employees

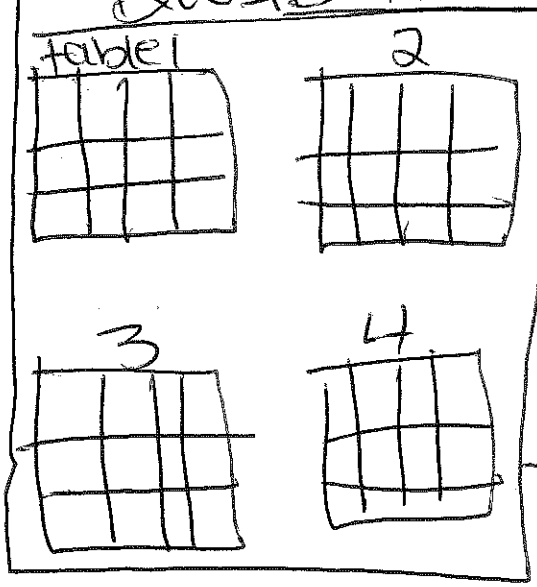
SELECT

24/06/26

SQL - JOINS

In a data warehouse we have multiple tables which are related to one another

DW - Data warehouse (relational database)



Database - place where we store data in tables

↳ Central data store

↳ A data warehouse have multiple databases

DBMS (e.g Databricks) - data management system

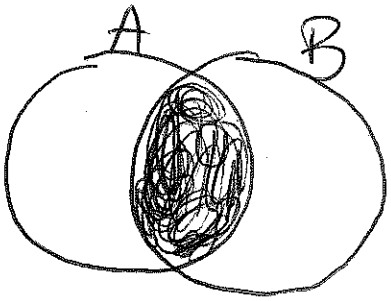
SQL coding

we can join multiple ~~tables~~ tables into One table

SQL Joins

↳ An SQL join is meant to join multiple tables based on common columns

INNER JOIN



We Join tables on columns but the Join affect the output rows.

-When we join table we can select any column between the tables we are joining

INNER JOIN_Returns rows where there is a match on both the tables.

Syntax: SELECT A.column1,
A.column2,
B.column3,
B.column4
FROM table1 AS A
INNER JOIN table2 AS B
ON A.common_column = B.common_column

Cust Id is a Primary key in the customers table (unique identifier)

Customers table

Cust_id	name
1	Rofhi
2 X	Mbali

Dimension table

Primary Key Orders

Order_id	Cust_id	Amount
101	1	500
102	4 X	20
103	5 X	10
104	1	300 new

Cust_id unique for each cust

Cust_id is common on both tables

not unique. Foreign key can be repeated

Fact table

SELECT A. Cust_id,
A. name,
B. Amount
FROM Customers AS A
INNER JOIN orders AS B
ON A.cust_id = B.cust. id;

Cust_Id	name	Amount
1	Rethi	500
1	Rethi	300

Bcc Cust_Id appears twice

new! now Rethi appears twice

inner joint - we want the thing that appers in both tables. = Outcome

Primary key on a specific table. Cust Id is a primary key

Customer Id is a foreign key in the orders table. Cust Id is not unique on Orders table, It can be repeated. Customer purchasing more than once - Id number didn't change. only order number changed.

Cust_Id gets created when you are Creating your Profile online (eg UberEats) (customer table)

Dimension table > Products table

Fact table - numerical table (orders table)

Products table (descriptive table)

Product_Id	~~Apple~~ Prod_name
131	Apple
132	Mango
133	Orange

product name doesn't change. Does not consistently change

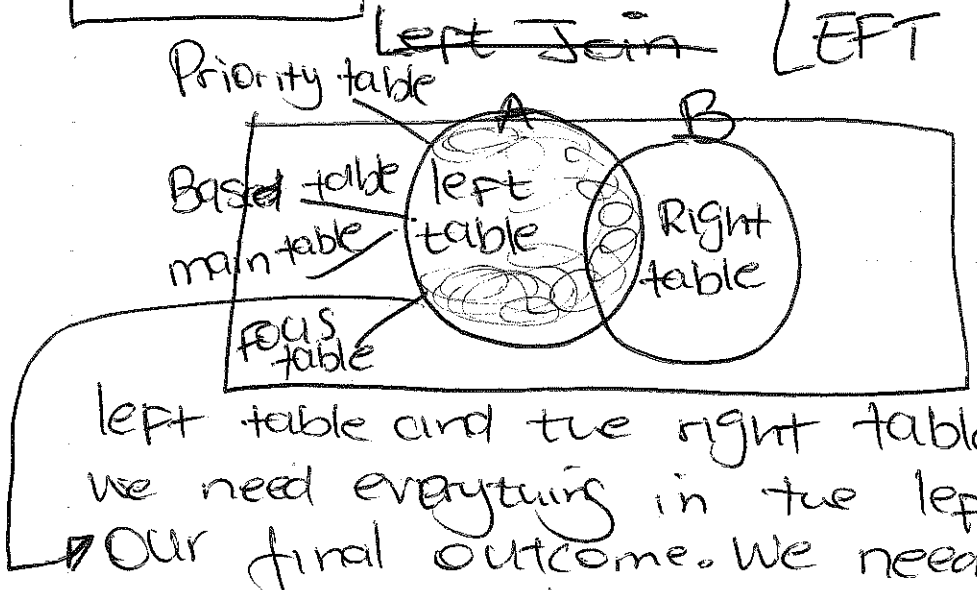
Dim table — Static descriptive information
↳ Data stored in a Dim table does not change frequently

Fact table contains numeric use for analysis.

oder_id	cust_id	Amount	Prod_id
101	1	500	131
102	4	20	132
103	5	10	131
104	1	300	133

24/06/26

~~Left Join~~ LEFT JOIN



left table and the right table
We need everything in the left table - In our final outcome. We need all rows in the left table

LEFT table is our Priority table
Contains critical information

From the right table - I will only take things that matches.

Syntax

~~FRO~~ SELECT column1,
column2,
column3,

FROM table1 AS A — left table
LEFT JOIN table2 ~~AB~~ AS B — right table
ON A.column_ Column = B.column_column

A LEFT JOIN returns all records from the left table, and only the matching records from the right table

When there is no match in the right table we still keep the left table row, but the right table colums will show NULL

Customers_table

Cust_id	Cust_name
1	Rofhiwa
2	Lerato
3	Mbali

Orders_table

Order_id	Cust_id	Amount
101	1	500
102	2	300

Write an SQL query to give me a list of all customers and what they Ordered.
— LEFT ~~JOI~~ ~~LEF~~ table is a priority (table A)

SELECT ~~A~~.cust_name,
~~A~~ Cust_id,
amount, order_id — left Table
FROM customers_table AS A
LETF JOIN orders_table AS B
ON A.cust_id = B.cust_id (joining 2 tables on cust Id)

(customer's table). Our main table

Cust_id	Cust_name	Amount	Order_id
1	Rethi	500	101
2	Lerato	300	102
3	Mbali	NULL	NULL

we are including Mbali although we don't know how much she spent, she is not on order table. AS NULL (her Order_id = NULL)

We are including her because she is in the LEFT table - we return all the rows in LEFT table

---

29/06/26 cheat sheet SQL

explain aggregate functions with example

30/06/26
Coffee_Shop_Analysis

~~Sale~~ & transaction time - (what time of the day has more sales) & ~~store~~
Store location - which store has more sales
Product category - which Product has more sales & Product type

Solve the Problems

1. SELECT *
~~WHERE~~ FROM coffe_table
WHERE store_location 'Manhattan'

~~* How many Products do we have~~

Determine the number of sales ~~per~~ per Product category and Product type in each Store from highest to lowest (DESC)

```
SELECT Product category,
       Product type
HAVING Sum

FROM table name((coffee_table)

WH·
```

```
SELECT_location,
       Product_category,
       Product_type,
       COUNT(*) AS total_transactions

FROM coffeeshop.brightcoffeeshop_bright_coffee_shop
GROUP BY store_location, product_category,
 product_type

ORDER BY total_transactions DESC
```

~~Start of notes~~

same table. Use CASE statement

```
SELECT
    store_location,
    CASE
        WHEN product_category IN('coffee',
        'Drinking chocolate') THEN 'Beverage'
        ELSE 'merchandise'
    END AS Product_classification,
    COUNT(*) AS total_transactions

FROM table name
GROUP BY store_location,
         Product classification

ORDER BY store_location, total_transactions
DESC
```

Select Statement is used to work with columns and ~~columns~~ to specify what columns you want to see in your output

Select * when you select everything

Semi-colon (;) is used at the end of the query

~~From~~ when you are working with more than 1 database on From, you need to specify the dataset and table that you are running the query from

E.g FROM parks_and_recreation.employee_demographics

(dataset name, table name)

when you are listing things you use a coma.

```
SELECT first_name,
       last_name,
       birth_date,
       age
       age+10 -> written to add 10 years to the age
FROM parks_and_recreation.employee_demographics;
```

PEMDAS (order in which the calculations are going to run in execution engine)
for comments

P_ Parentheses ( )
E - Exponents - Power Functions. ^
MD - Multiplication & Division - share the same level of Priority.
AS - Addition + Subtraction - Also Share the level of Priority - moving from Left to right

DISTINCT e.g if you have a table with females & males : SELECT DISTINCT gender
- the output will show only 2 rows which is male & female

Best Graphs options

Bar Chart

- To show viewers count by Province, gender, or race.
- Can use color to add a second dimension (e.g bars by Province, coloured by gender)
- Best for comparing totals across categories

Bright_case_study _case study
_viewership

LEFT JOIN

miro - create an account
1. `google

1. User Profile
first sheet - ~~viewership~~ viewership

table name - user _profiles
Upload them separately

Case study 01/07/2026

Bright TV

Grow the company's subscription base
what must we do to attract ~~Subs~~ viewers
Customers

Retain customers

e.g Netflix writing or offering discounts -
they use data to see or know your
patterns

CVM (Customer Value Management)

Data driven questions:

Provide insights - information that comes
from the data

Factors that influence consumption - e.g
during wordcup - people watch DSTV alo

Actionable insights - take information &
take action
whatever that
comes from data
must be Actionable

Notes:

(Business requirement)
Problem we are solving - Low customers

Miro reading &
Step 0: ~~Data~~ Understand business requirment

Step 1: Asking Data Driven Questions

lookalike customers —

Step 2: Data inspection
ingest the Data on databricks
↓
checking the available data

Step 3: Data processing
↓
After inspecting Data, Check what can be done from the data.
This includes data cleaning, data manipulation

---

bright viewership sheet 02/04/26
bright User profile

ELSE - IF gender is not unknown - Give me the gender
ELSE - gender

Province

Select ~~distinct~~ DISTINCT ~~Province~~

CASE

WHEN Province = 'None' THEN 'Unknown'
WHEN Province = ' ' THEN 'Unknown'
WHEN Province IS 'NULL' THEN 'Unknown'
ELSE Province ← new column
END AS ~~Province~~ Region

END AS Region

Region

Region
~~Free State~~
EC
KZN
GP
WC
NW
LP
MP
NC
Unkown

Age ~~bracket~~ check

SELECT Min(Age) AS min_age — To check youngest person
Max(Age) AS max_age — To Find the oldest person
FROM user_profile AVG(Age) AS mean_age
'around age, not exact age btwn lower & above

```
SELECT
    CASE
    WHEN Age = 0 THEN 'infant'
    WHEN Age BETWEEN 1 AND 12 THEN
      'kids'
    WHEN Age BETWEEN 18 AND 35
    THEN 'YOUNG Adults'
      13 AND 17 THEN 'Youth'
      35 AND 50 'Adults
    Age > 50 THEN AND Age <= 60
    THEN 'Elder'
WHEN Age > 60 THEN 'Pensioner'
END AS Age_group

FROM  User_Profile

    RUN ALL - google it
```

and Data manipulation

Step 4: ~~canva~~ canva
Snowflake

Step 5: Data Processing

Miro - Add-on User Profile

Are the Customers contactable?

After inspecting the Data, check what can be done from the Data. This include Data ~~Cleaning~~ E

Can I Personalize the campaigns to send to the Customer?

Step 3: Data Inspection

~~ETL~~

Ingest the Data on Databricks

checking the available Data

Steps to clean the data - On the Project (Case Study) Bright TV

User profile

1. Inspect data – SQL Editor

```
SELECT *
FROM User_Profile
Limit 100;
```

Run the code

```
SELECT DISTINCT gender
FROM User_Profiles;
```

~~Gender~~
Gender
Male
Female
None
blank

output

```
SELECT DISTINCT
    CASE
    -When gender = 'None' THEN 'UNKOWN'
    -WHEN gender='' THEN 'Unknown'
    -WHEN gender IS NULL THEN 'Unkown'
    -ELSE gender
-END AS ~~gender~~
-FROM User_profiles;
```

14/07/26

N-operator to join more than One column.

Part-to-whole breakdown e.g out of 100 People how many are black People and how many are white People
-use a stacked bar chart or a pie chart (< 6 categories or options)

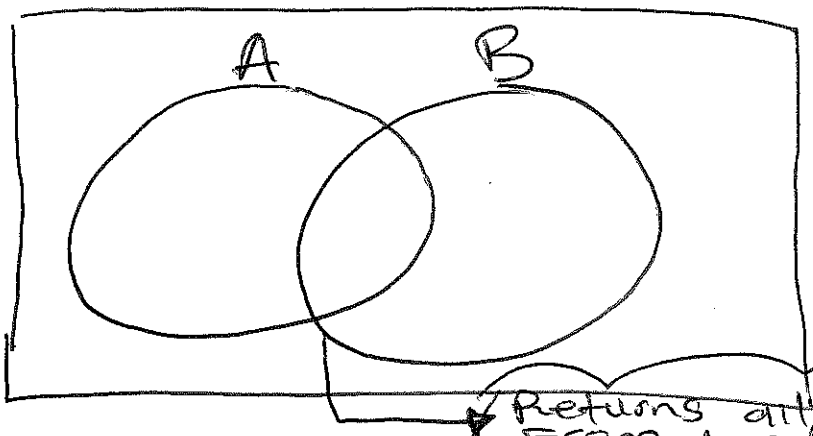
Page 1

2 distinct count of sub-H
Ethnicity Regions

time trends
how to round off hours

Values
Sum of duration hours
Rows Hour_of_day

LEFT JOIN



Returns all the rows from the left table & only the matching rows from the right table

```
SELECT col1,
       col2,
       col3,          ← Left table
FROM table1 AS A
LEFT JOIN table2 AS B  ← right table
ON A.common_column = B.common_column
```

You can JOIN more than one table/columns that are common.

Customers

Customer	Cust ID	CustName
	1	Thabo
	2	Sarah
	3	Lerato
	4	John

Orders

```
SELECT Customer ID,
       Customer Name,
       Products
FROM Customers AS A
LEFT JOIN orders AS B
ON A.customer ID = B.customer ID
```

Customers

Cust ID	Cust Name
1	Thabo
2	Sarah
3	Lerato
4	John

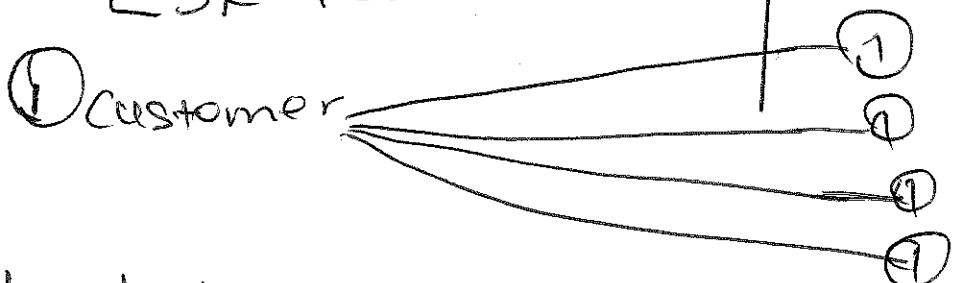
Orders

Order ID	Cust ID	Product
101	1	Laptop
102	2	Phone
103	2	Headphones
104	4	Tablet

Output (Previous page code)

Cust ID	Cust Name	Product
1	Thabo	Laptop
2	Sarah	Phone
2	Sarah	Headphones
3	Lerato	NULL
4	John	NULL

Dimension Table	Fact Table
Descriptive Table	Analysis
Stat	Continously change
	↓
User_Profile	Viewership
5K rows	



You decide which is the LEFT table, ALL information on the LEFT table will be selected. Bare in mind when you choose your JOINS
- which table contains the main/~~main~~ alot of information

Join the table with least information to the one that has alot of information.
LEFT Table is the one with alot of informa

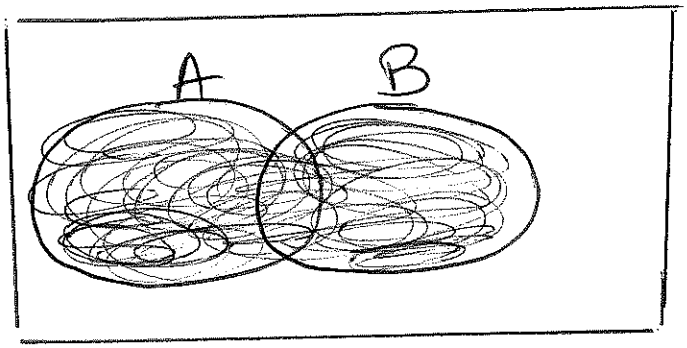
15/07/26

FULL OUTER JOIN

A FULL OUTER JOIN returns;

- All matching rows
  - All the rows that are matching on the common columns are combined into a single row

- All Unmatched rows
  - where there's no match, we will put a NULL



Syntax:

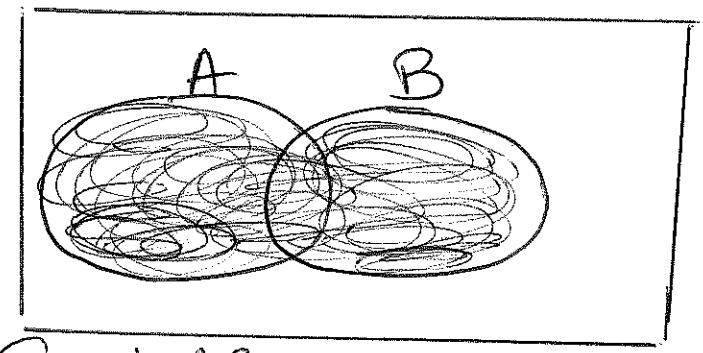
```
SELECT col1, col2, col3
FROM table1 AS A
FULL OUTER JOIN table2 AS B
ON A.common_column = B.common_column,
```

Join the table with least information to the one that has alot of information.
LEFT Table is the one with alot of inform

15/07/26

FULL OUTER JOIN

A FULL OUTER JOIN returns;
- All matching rows
  (All the rows that are matching on the common columns are combined into a single row
- All unmatched rows
  (where there's no match, we will put a NULL



Syntax:
```
SELECT col1, col2, col3
FROM table1 AS A
FULL OUTER JOIN table2 AS B
ON A.common_column = B.common_column;
```

Customers

Cust ID	CustName
1	Thabo
2	Sarah
2	Lerato
3	
4	

orders

OrderID	CustID	Product

```
SELECT  CustumerID
        CustomerName,
        OrderID,
        Product
FROM customers AS C
FULL OUTER JOIN orders AS O
ON C.customerID = O.customerID;
```

Output

CustomerID	CustomerName	OrderID	Product
1	Thabo	101	Laptop
2	Sarah	102	Phone
2	Sarah	103	Headphones
3	Lerato	NULL	NULL
4	John	NULL	NULL
5	NULL	104	Tablet

we don't know what Lerato ordered

we do not know the name of customer number 5, we only know their OrderID & that they bought a tablet.

UNION	UNION ALL
- removes duplicates	- does not remove duplicates
Combine the results of two SELECT statements by stacking the rows	

- ~~There Each~~
- The Select Statements must have the same number of columns
- The colums must be in the same order
- The columns must have the same data types.

current_students

Student_Id	StudentName
1	Thabo
2	Sarah
3	Lerato

New_students

Std_Id	StdName
3	Lerato
4	John
5	Marry

```
SELECT student_Id, StdName
FROM current_students
UNION
SELECT std Id,
       StdName
FROM New_students
```

Student_Id	StudentName
1	Thabo
2	Sarah
3	Lerato
4	John
5	Marry

Order of matters on std Id & stdname when writing the Syntax from both tables when mentioning the columns.

UNION ALL

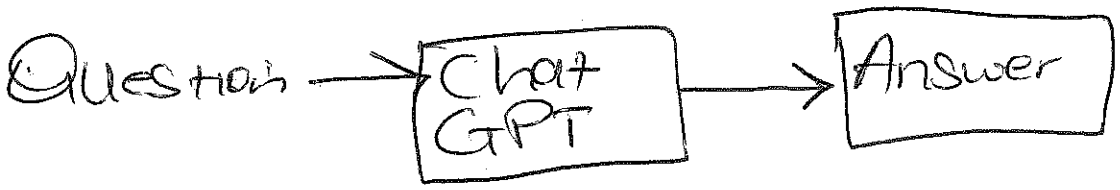
Student	StudentName	
1	Thabo	Lerato appears twice because UNION ALL does not remove duplicates.
2	Sarah	
3	Lerato	
4	John	
5	Marry	
3	Lerato	

Sequence on duplicates, does not matter 'Lerato is a row = sequence does not matter.

Data AI AGENT 17/07/26

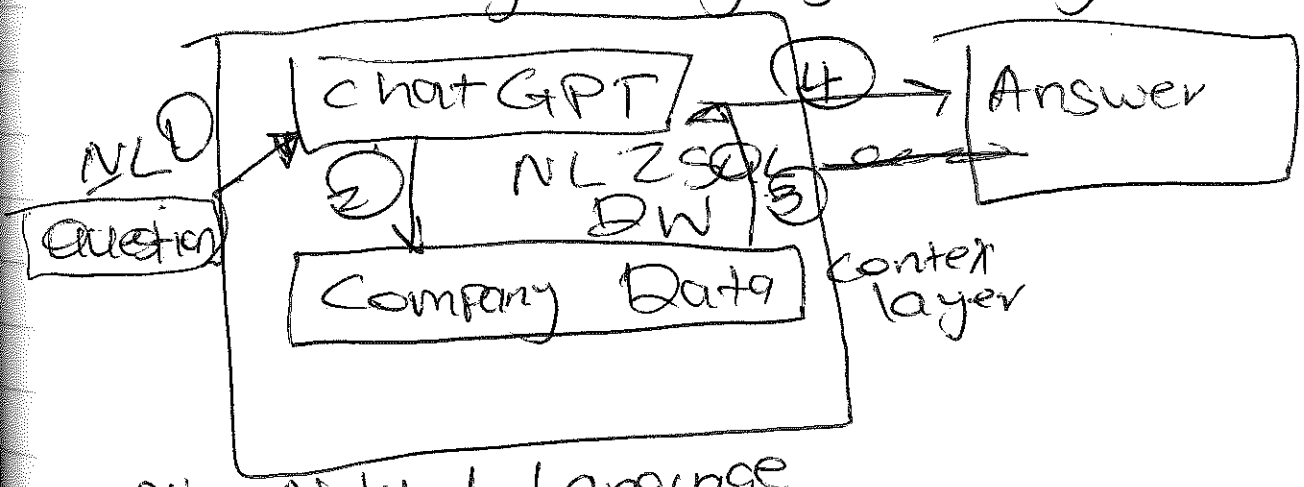
What is a data agent

You take a code/question to CHATGPT OR LLM
You use natural language



ChatGPT
- does not known data in a given company OR internal company data.

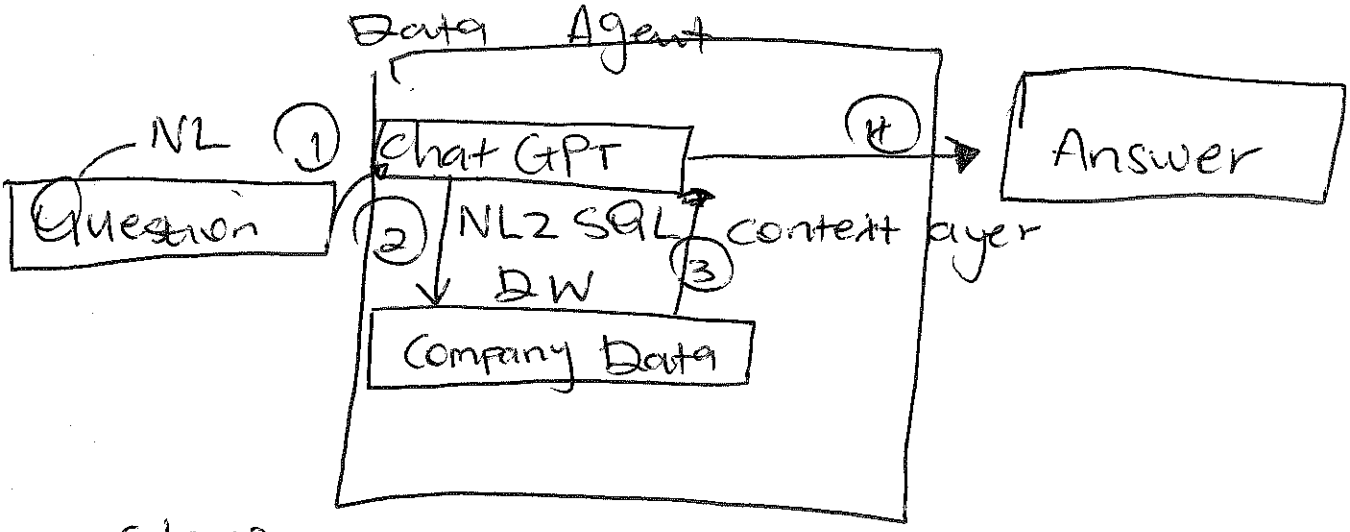
LLM - Large Language Managemet.



NL = Natural Language

When we have a Data agent, we ask it a question in Natural Language. The question gets converted into SQL, that SQL script ~~it runs~~ - we query that SQL against the database, we retrieve the output and give it back to the LLM

to do the analysis for us and send back complete analysis.



Steps:

1. On databricks go to Genie Agents
2. +(Add) New
2. Connect to your data
   Choose dataset
   Create (A chat will be created for you to ask questions)

e.g What can you do for me?
It will analyse data for me

Second question:
What are the different store location and how much is each store making in a month?

Go to configure - top right of the screen
- Sources

It took the Natural Language (NL) and converted it into SQL Script

question 3:
Show me a donut chart that gives revenue distribution across three store locations and the top 5 performing products.
Show code
check or double check the code and correct it IF there are mistakes.

Data Agent Instructions - General instructions
- Add general instructions on how you want Genie Agent to behave - Under Configure

- Add examples
Add- example query
- Filter
- Measure
- Field
- Join
write a SQL script and it show it examples

ALL under Configure

Also sources
- At Add all your different tables as Sources - You can connect mutiple tables

On ABOUT - add a description e.g Coffee Shop business Analyst which Provides insights or analysis on the bright coffee shop and also gives recommendations on actions to take in order to grow the business.

Edit agent details
Name: coffee Shop business Analysis

If you want to key: value

If you want to add a picture - you can on thumbnail

Ground truth

Fact based questions

NO need.

microsoft foundry - multi agent workflow - Orchestrator Agent — we will do it later

Data portfolio

- Miro — what did you do on Miro
  SQL
- Design presentation & analytics
- Data story telling on canva
- Design project plan on Canva to track the progress of analytics project

Data analyst student at Brightlearn
— Dates.
work intergrated learning

Creating Created a databricks to GitHub intergration —

GitHub

or portal
Go on Brightlearn website to check skills — data analytics
— Download the study guide

Also use chatGPT or claude
1. Use Free AI TOOLS to build your portfolio
   — Brightlearn portal.
2. Add Vibe coding

3. Data agent guide — All under additional documents.
4. SQL master Guide.
   upload the docs on CHATGPT or claude & write a PROMPT (question) & add your CV

AI website & APP builders

1. LOVEABLE
2. V0 by Vercel
3. Bolt.new
4. Claude Artifacts

AI coding & writing helpers

~~1. ChatGPT~~
2. Google Gemini
3. GitHub copilot
4. Cursor

AI Design tools

1. canva 2. Figma 3. Framer 4. Microsoft Designer

Free Hosting & Development

1. Netlify
2. Vercel
3. GitHub
4. cloudflare pages

1. Plan & write your content
   - Upload study guide on chatGPT & ask prompt - extract info & put it in a nicer way.

2. Generate the site or copy
   - Use LOVEABLE - take prompt from ChatGPT or Claude to LOVEABLE - & paste it there.

3. Polish the visuals
   - logo & project thumbnails - in canva or microsoft designer.

~~First step~~

1. Upload Brightlearn guidelines/documents relevant to data ~~analytics~~ analytics on chatGPT or claude
2. Upload your CV - on claude/ChatGPT
   ", write a prompt for me to create a ~~afford~~ modern digital CV portfolio using LOVEABLE indicating all the skills that I have learnt and the skills that I ~~abtre~~ already had.

You can also use Replit
base44

~~bl~~

#Backticks — if we select columns with no underscore e.g province clean

#Brackets — stand alone query statement that says one query

#Comas — listing colums

Selecting — something in the rows - useful words use single quotes e.g 'Graduteng' '---

Do more research on this

research on sub query - how to choose ~~you ca~~ between CTE and sub query.

You can use a CTE even when working with one table - Do research!!

CTE starts with a 'WITH' we are cleaning the specific table e.g ~~*WITH~~ WITH user profile (we are cleaning this table)

write 'Email before email cleaning & Social media cleaning part

Data visualization bootcamp

Depicting info about data in to a non-technical people

Summarizing observation from our data

Presenting data in the form of graphs, charts, trends

DATA VISUALISation tells a story faster

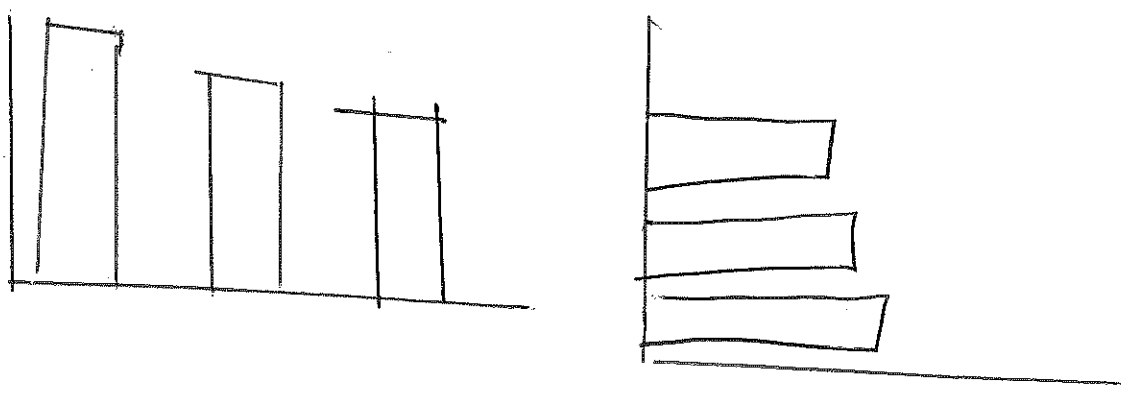
→ When to use which graph

A graph always answers a question
What question are you trying to answer

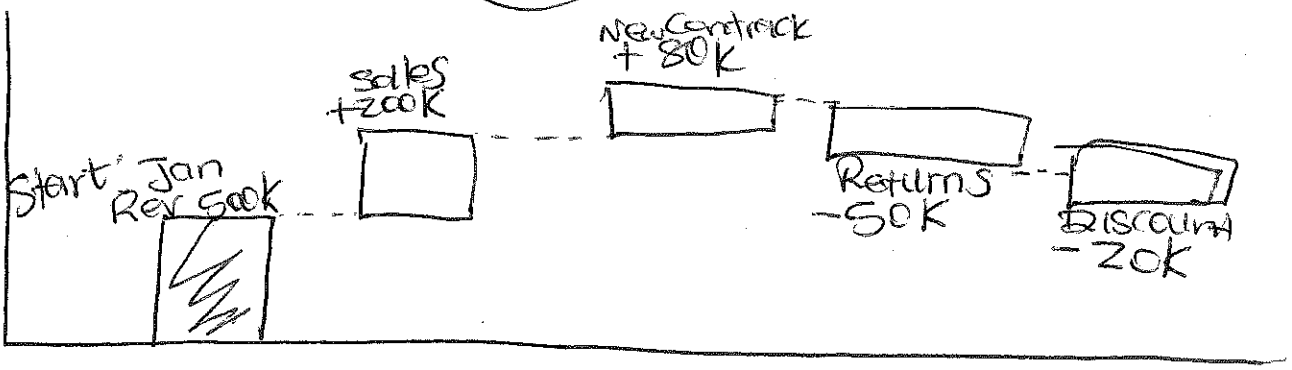
↳ But which graph do we use ≠ when?

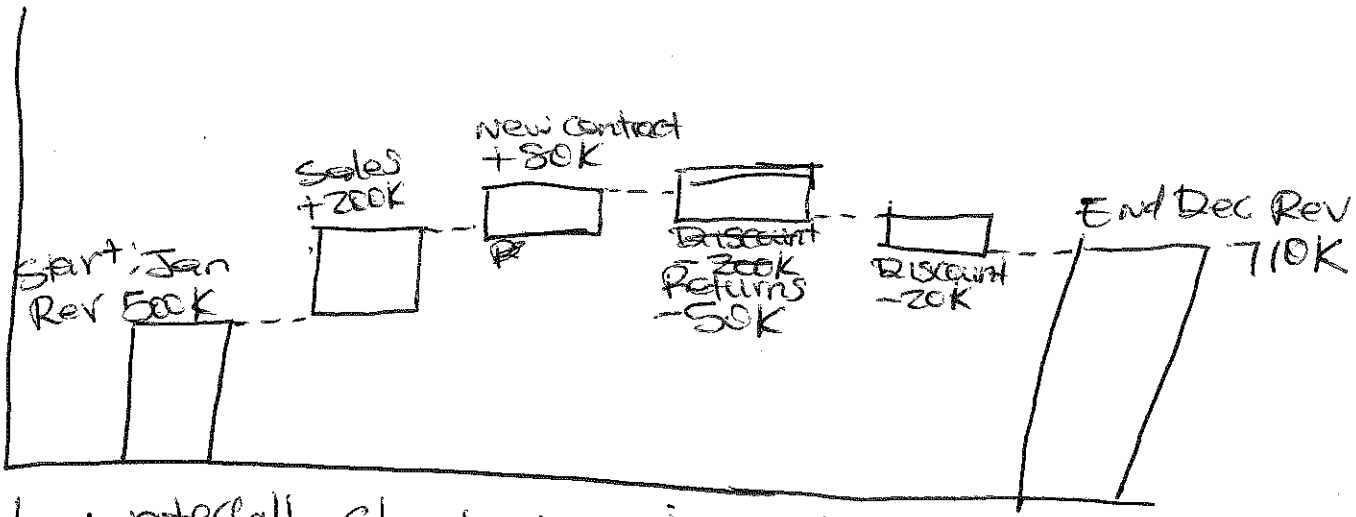
Bar graphs

For comparing categories
- Ranking items of showing differences between groups



Waterfall Chart



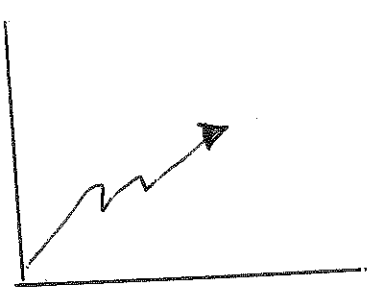


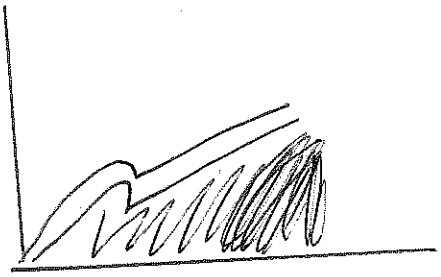
Lo waterfall chart visualises the impact of positive and negative changes to explain how you got from starting point to ending point.

Line graphs

For tracking Time & trends

- Showing how something changes over days, months or years

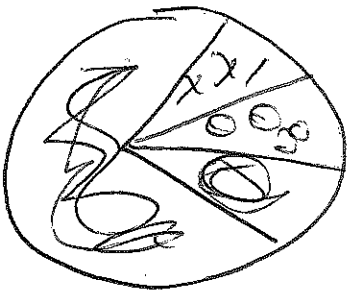


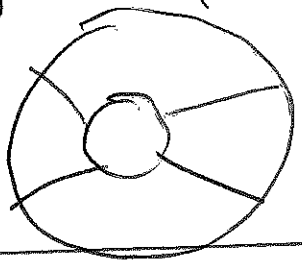


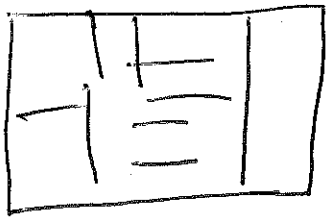
Pie charts

Showing ~~parters~~ Proportions

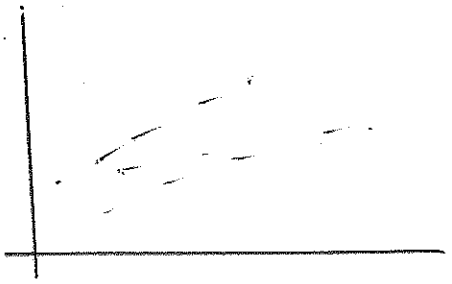
- Illustrate how parts contribute to a whole (e.g out of 100%)

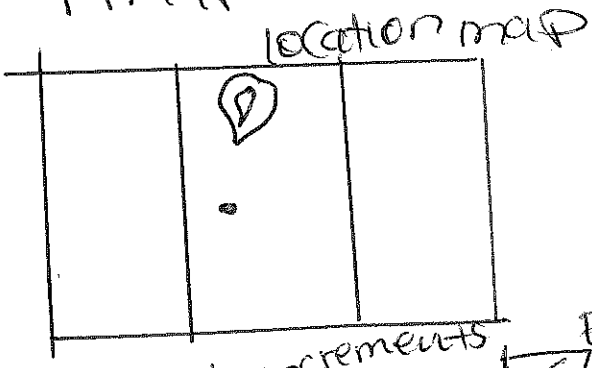




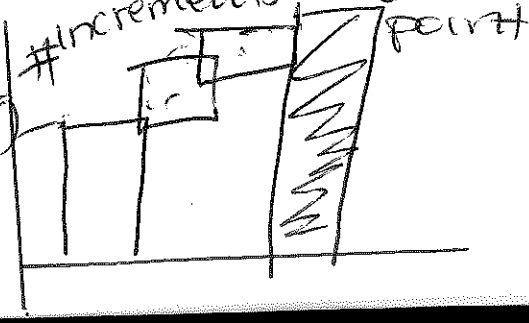


Scatter ~~Plots~~ + maps





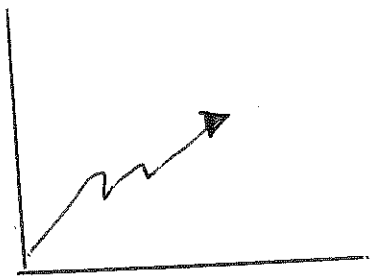
- Exploring relationships, geographical patterns or step-by-step financial changes

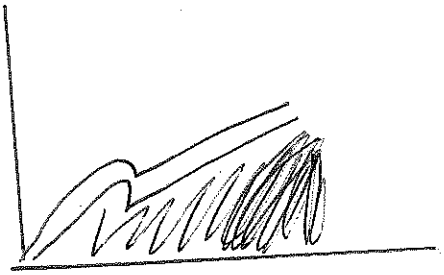


Line graphs

For tracking Time & trends

- Showing how something changes over days, months or years

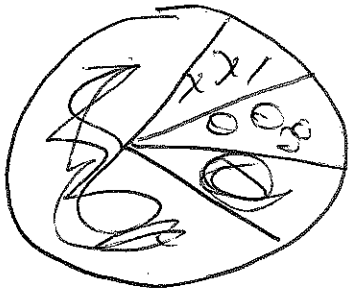


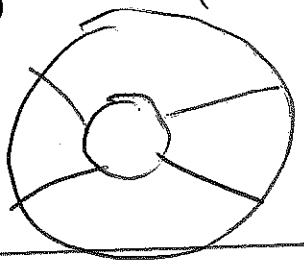


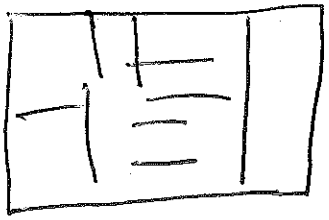
Pie charts

Showing ~~portions~~ Proportions

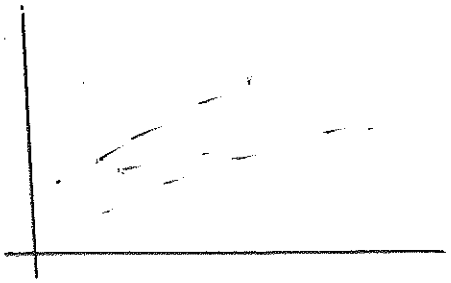
- Illustrate how parts contribute to a whole (e.g out of 100%)

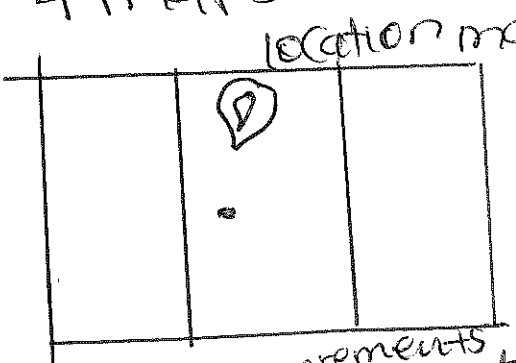




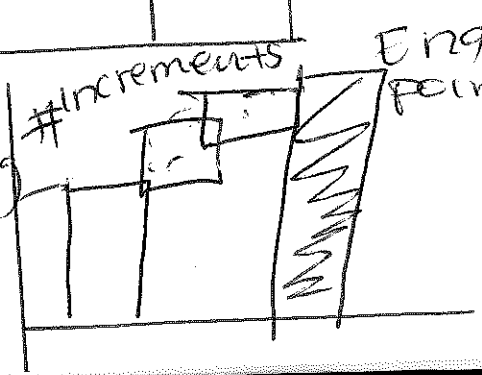


Scatter plots + maps

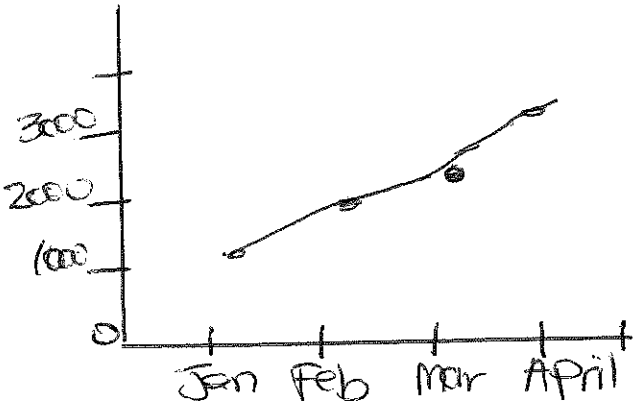




- Exploring relationships, geographical patterns or step-by-step financial changes



Line chart

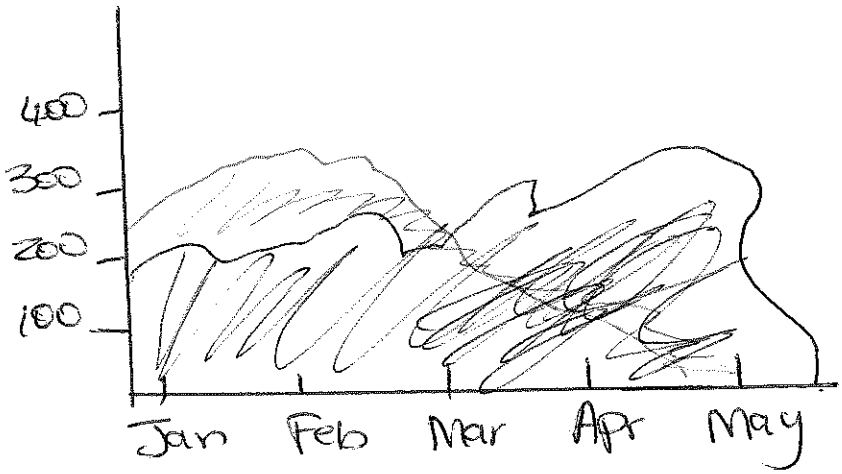


*When to use it:
when displaying continuous trends over time
longer periods

* In the above example, it shows steady growth from Jan.–April.

* TIP: Sequence matters!! Use the data points in chronological order.

Area chart

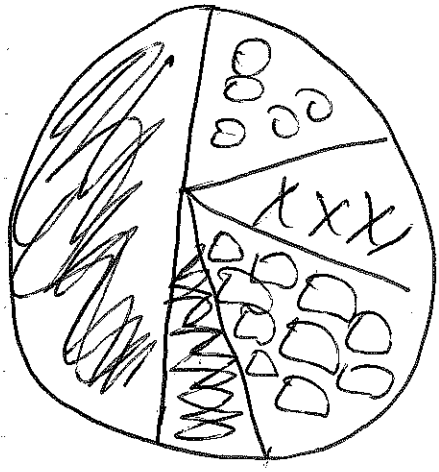


When to use?
When highlighting the magnitude of change, not just the trend.

-This example shows the power utility usage of 2 provinces from Jan-May.

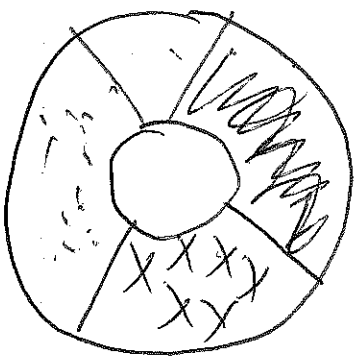
* TIPS: Use this graph to show how different parts contribute to the total over time.

Pie chart



TIP: Maximum 6 slices
More than 6 slices ~~beg~~ becomes difficult to read

Donut chart



TIP: * Mostly used in dashboards for a modern look & space efficiency.

*Stick to no more than 6 slices.

Treemap

Office space occupancy by department

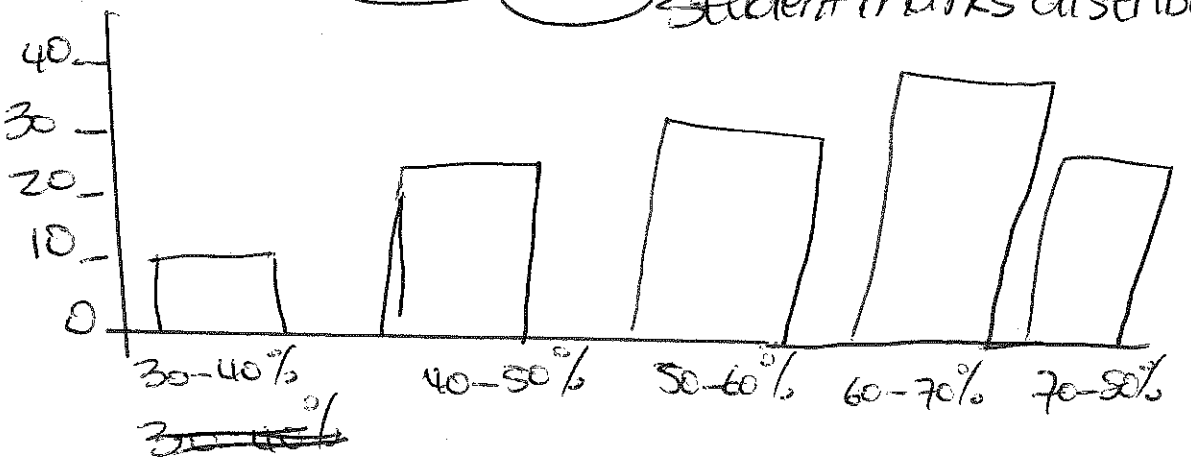
Finance 25%	marketing 30%	
IT 15%	Operations 10%	Sales 10%
AP 10%	Procument 7%	Exco 5%

* TIP: use this ~~group~~ graph when you have too many categories to fit into a pie chart or donut.

(*show top 10 - bar graph (best chart))

Histogram

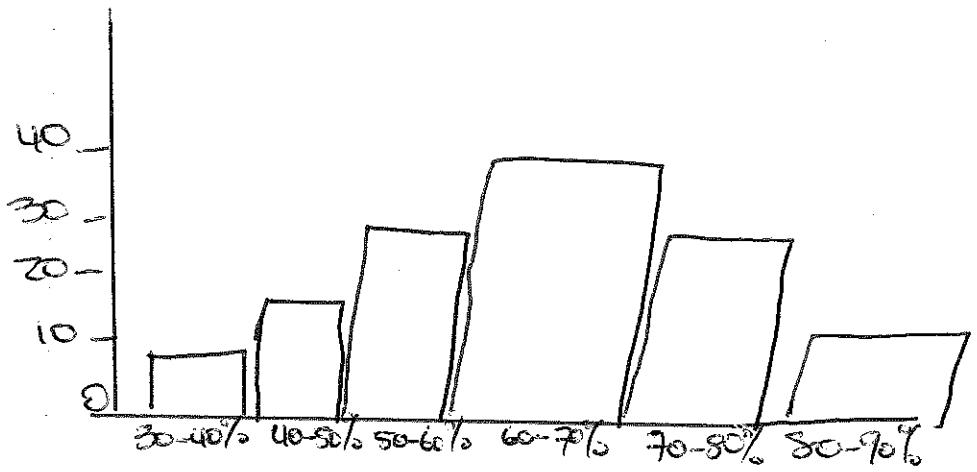
~~student~~ marks distribution



~~30-40%~~

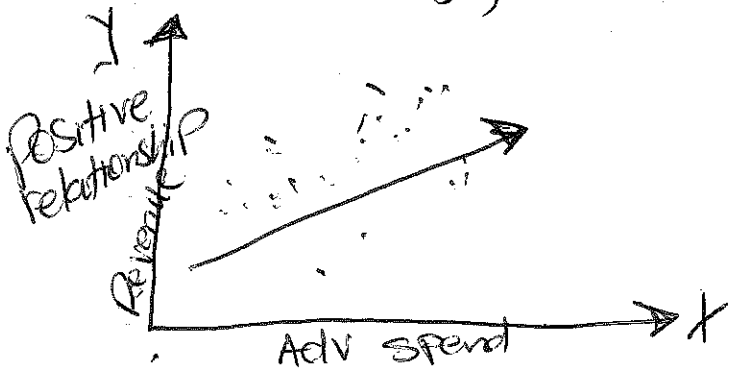
~~Histogram works~~

Lo Histograms works best to show intervals
Lo From the example, students who got 30-40 are 10

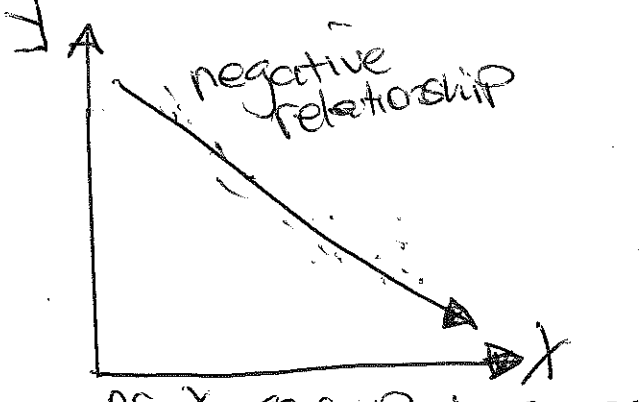


Scatter plots

used for ~~exploring~~ relationships between two numeric variables & spotting unusual data points (outliners)



As X goes up, Y goes up
E.g higher spend on advertising leads to higher sales



As X goes up, Y comes down

